Survey and Contamination

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Contents

1 Objective

To practically assess ambient radiation dose rates and quantify radioactive surface contamination within a clinical radioisotope facility.

2 Apparatus

- Portable Ionization Chamber Survey Meter
- Geiger-Müller (GM) Contamination Monitor
- Standardized Check Sources for functional verification
- Personal Protective Equipment (PPE) and smear test wipes

3 Theory

Nuclear medicine departments handle unsealed radionuclides for diagnostic imaging and targeted therapy. Unlike sealed-source applications, the radiopharmaceutical can spread outside the vial through preparation errors, spills, aerosols, patient excreta, contaminated disposables, and poor workflow segregation. Therefore, *radiation survey* and *contamination monitoring* are not supplementary tasks but foundational radiation protection operations that support safe daily clinical practice [1, 2, 3].

3.1 Protection Philosophy and Regulatory Basis

The theoretical basis of this experiment is rooted in three internationally accepted principles:

- **Justification:** a medical radiation procedure should provide net benefit;
- **Optimization (ALARA):** exposure is kept as low as reasonably achievable considering social and economic factors;
- **Dose limitation for workers/public:** systematic controls are required for non-patient exposures.

These principles are articulated by ICRP and implemented in IAEA safety requirements for medical uses of ionizing radiation [4, 5, 2]. In India, institutional implementation is aligned with AERB's regulatory framework and the Atomic Energy (Radiation Protection) Rules [3, 6, 7].

For a nuclear medicine unit, this translates into a practical question: *Is radioactivity staying where it is intended to be?* Area survey answers the external field component, while contamination survey answers the source control component.

3.2 Concept of Area Survey

Area survey is the measurement of ambient radiation field in and around working zones. Typical monitored points include dose preparation benches, dispensing counters, syringe transport routes, uptake rooms, imaging gantry surroundings, patient toilets, waste decay storage, and nearby unrestricted corridors.

From a physics perspective, measured quantity is usually ambient dose equivalent rate (or an operationally equivalent meter reading), reported in μSv/h. Repeated measurements over time establish a baseline map. Deviations from baseline are often more informative than a single reading because they indicate process drift, unexpected source accumulation, or shielding compromise [2, 8].

Area survey data are used to:

- classify and verify controlled/supervised regions,
- optimize occupancy and staff traffic routes,
- validate shielding assumptions in routine operation, and
- detect unusual source retention (for example in waste holding or uptake areas).

3.3 Concept of Contamination Survey

Contamination monitoring addresses the spatial transfer of unsealed activity from intended containment surfaces to unintended surfaces. In practice, contamination is classified as:

- **Fixed contamination:** activity strongly bound to the surface, difficult to remove by gentle wiping;
- **Removable contamination:** activity transferred to gloves/swipes and therefore immediately relevant for spread and internal exposure pathways.

The key radiological concern of removable contamination is pathway formation. Once transferred to skin, clothing, handles, phones, records, or food-contact surfaces, exposure risk shifts from external-only to combined external plus potential internal intake (ingestion/inhalation). Hence removable contamination has high operational significance even when total activity appears small [1, 2].

3.4 Detector Physics Relevant to the Experiment

3.4.1 GM-Based Contamination Monitoring

Geiger-Muller detectors are widely used for contamination checks because of high sensitivity and fast response. In Geiger operation, one initiating ionization event triggers a full avalanche; pulse amplitude is nearly independent of deposited energy. Therefore, count rate is suitable for detection and screening but not for precise dosimetry. Important limitations include:

- dead-time losses at high count rates,
- energy dependence,
- geometry dependence for extended/non-uniform contamination,
- limited radionuclide discrimination.

These effects must be considered when interpreting high-count situations near concentrated activity [9].

3.4.2 Ionization-Chamber Area Survey

Ionization chamber meters operate in the saturation region with no gas multiplication, so signal is proportional to ion pairs created in the chamber volume. This gives improved dose-rate fidelity and better behavior in moderate-to-high field conditions compared with GM counters. Hence ion chambers are preferred for quantitative area radiation survey, while GM meters remain practical for contamination tracing [10, 8].

3.5 Mathematical Basis for Wipe-Test Quantification

For removable contamination, a known area S is wiped and counted. If measured count rate is C_{gross} and background is C_{bg} (in CPM), then

$$C_{\text{net}} = C_{\text{gross}} - C_{\text{bg}}. \tag{1}$$

Estimated removable activity on the wipe is:

$$A_{\text{rem}} = \frac{C_{\text{net}}}{60\,\epsilon\,f}, \tag{2}$$

where ϵ is counting efficiency and f is wipe recovery fraction.

Area-normalized removable contamination is:

$$a_{\text{rem}} = \frac{A_{\text{rem}}}{S} \quad (\text{Bq/cm}^2). \tag{3}$$

If counting is delayed by Δt after sampling for radionuclide decay constant λ, the decay-corrected activity at sampling time is:

$$A_0 = A_{\text{rem}} e^{\lambda \Delta t}. \tag{4}$$

These relationships show that contamination result quality depends not only on detector response but also on wipe technique, counting geometry, radionuclide calibration, and timing control [1, 9, 2].

3.6 Minimum Detectable Activity and Practical Sensitivity

In low-level contamination work, non-detection does not imply zero activity. The practical statement is that activity is below the system's detection capability under given counting time and background. A longer counting time, optimized detector-window selection, and stable background can significantly lower the detectable threshold. Therefore, routine programs should keep measurement conditions consistent so trend comparison is meaningful across survey days [9, 2].

3.7 Survey Design in Nuclear Medicine Workflow

A scientifically sound survey program is risk-based rather than random. Sampling density is highest at transfer points where activity changes hands or containers. Typical priority nodes are:

- dose drawing and assay bench,
- syringe shielding and transport trays,
- injection chairs and patient hand-contact surfaces,
- floor zones near waste bins and sink drains,
- patient toilets and release pathways,
- decay storage interfaces.

The same route should be repeated with a fixed sequence, probe distance, and counting dwell to reduce operator-dependent variability. Mapping data on a floor plan improves interpretability and supports corrective action prioritization [2, 7].

3.8 Response Logic for Contamination Events

When elevated contamination is found, response follows a control hierarchy:

1. isolate and label the affected area,
2. prevent spread through controlled movement and PPE,
3. perform staged decontamination,
4. re-survey after each stage,
5. record final status and probable root cause.

Repeated re-survey is essential because a single post-cleaning reading can be misleading due to residual wetness, detector positioning, or temporary redistribution. Event documentation should include location, radionuclide (if known), first reading, decontamination method, final reading, and preventive recommendation [2, 3].

3.9 Uncertainty Sources and Measurement Limitations

Theoretical interpretation of survey data requires explicit recognition of uncertainties:

- **Statistical fluctuation:** counting follows Poisson behavior, dominant at low count levels;
- **Instrument factors:** calibration status, dead time, battery condition, energy dependence;
- **Geometric factors:** probe-to-surface distance, orientation, and source non-uniformity;
- **Sampling factors:** wipe pressure, area definition, surface roughness, moisture, recovery factor;
- **Temporal factors:** decay during delay before counting.

Hence, reporting should avoid false precision. A good practice is to combine numeric results with categorical interpretation (background level, elevated but controlled, contamination event requiring intervention) aligned with institutional action criteria [9, 2].

3.10 Quality Assurance and Record-Based Safety

Radiation protection in nuclear medicine is fundamentally a record-driven control system. Trend plots of area dose rate and contamination results provide early-warning information before a major event occurs. QA elements include instrument constancy checks, periodic calibration traceability, defined survey frequency, documented action levels, and training refreshers for all handlers of unsealed sources [6, 5, 2].

In this context, the experiment demonstrates not only measurement technique but also the operational philosophy of nuclear medicine safety: detect early, intervene promptly, verify objectively, and document completely.

3.11 Summary of Theoretical Significance

This experiment integrates detector physics, contamination transfer mechanisms, and regulatory safety philosophy into one practical framework. Area survey quantifies external radiation fields; contamination survey verifies source confinement; swipe quantification links count data to activity metrics; and regulatory guidance ensures that measurements are translated into defensible safety actions. Together, these components form the scientific and operational basis for radiation protection in nuclear medicine departments [4, 5, 2, 3, 7].

4 Observation

Location	Radiation level(μR/h) Ion chmaber	Contamination level(CPS)	Radiation level(mR/h) GM based	Sample test(dpm)	Sample test(Bq or dps)	Remark
Dispensing room	15	0.53	0.06	0.00	0.00	Within tol
Dose administration room	10	0.63	0.02	193.80	3.23	Within tol
Post administration waiting	17	0.63	0.01	113.80	1.90	Within tol
Decontamination room	11	0.73	0.02	0.00	0.00	Within tol
Scan room	20	0.33	0.03	106.20	1.77	Within tol
Console	15	0.63	0.02	26.15	0.44	Within tol
Reporting room	15	0.33	0.01	153.80	2.56	Within tol
Restricted corridor	29	0.43	0.03	70.77	1.18	Within tol
Reception	12	0.73	0.03	89.23	1.49	Within tol
Nursing area	17	1.13	0.05	118.50	1.98	Within tol
Waste storage	23	0.73	0.04	43.08	0.72	Within tol
Source storage*	15	0.73	0.03	-	-	-

Tolerance Limits:

Sr. No.	Category of areas	Limits of surface contamination
1	Monitored areas (e.g., inside the fume hood)	$10^{-3} \mu Ci/cm^2$ ($37\ Bq/cm^2$)
2	Laboratory areas (Surveyed)	$10^{-4} \mu Ci/cm^2$ ($3.7\ Bq/cm^2$)
3	Other non-active areas	$10^{-5} \mu Ci/cm^2$ ($0.37\ Bq/cm^2$)

5 Conclusion

A comprehensive understanding of the governing physical principles and stringent Quality Assurance protocols ensures optimal clinical outcomes and fundamental radiation safety.

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